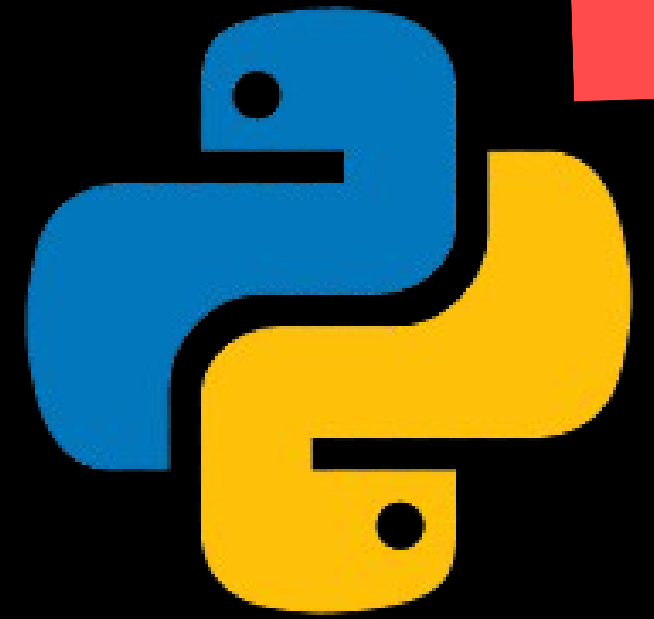


Video / Steps

1. Introduction and concept clarification
2. Installation
3. Bill Image cleaning
4. Text extraction from bill
5. Defining prompt and LLM (AI) for inputs from the bill & categorization
6. Creating table for storing data from bills
7. converting data extracted into dictionary format for insertion into table
8. Data moving from extracted text into the table (output from step 5 & 6)
8. Frontend creation using streamlite - (Phase I complete)
9. Creating LLM agent and prompt for conversation with data
10. Integrating fully functional LLM agent in the frontend

Talk with your expense bills!



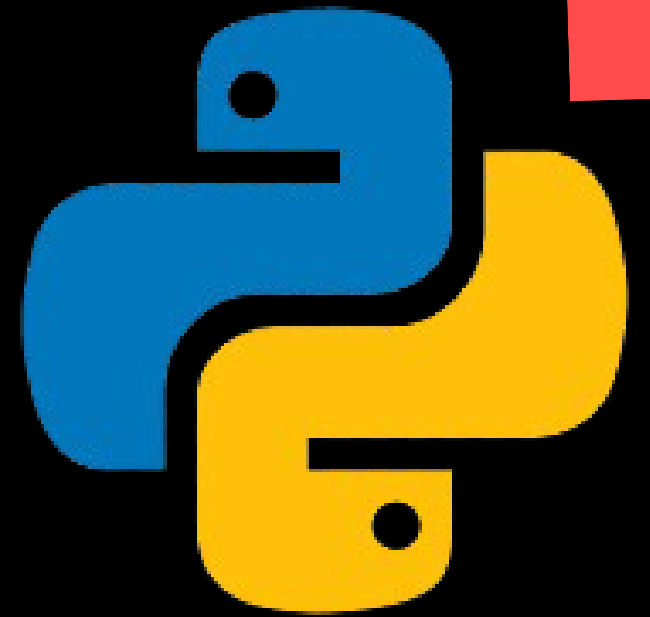
#1 Flowchart / Conceptualization

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#2 Installation

1. python version <= 3.12
2. VS studio
3. Ollama (LLM_AI)

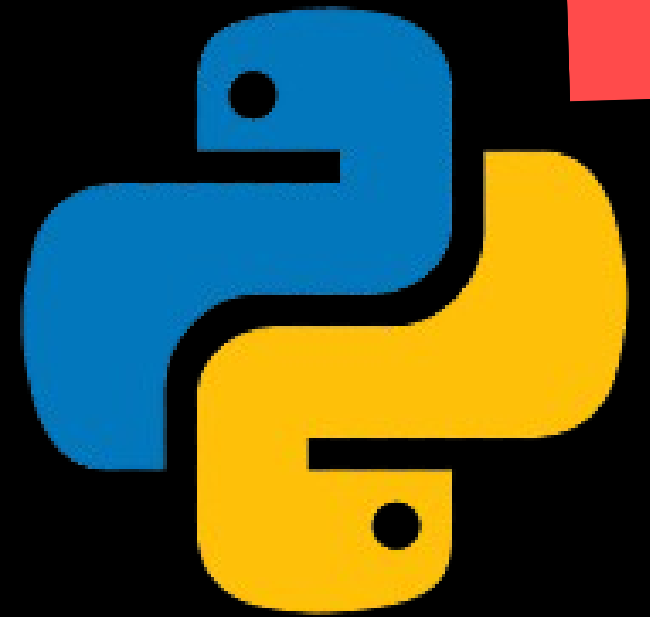


100

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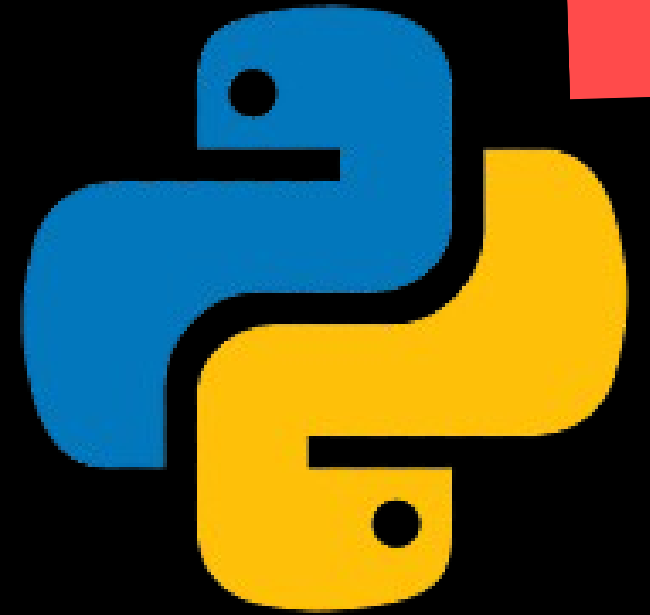


#2 Installation



Adding Ollama (AI) into Environmental Variables

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#3 Image Cleaning process

1. Convert image into grayscale.
2. Gaussain cleanup
3. Otsu's binarization of image

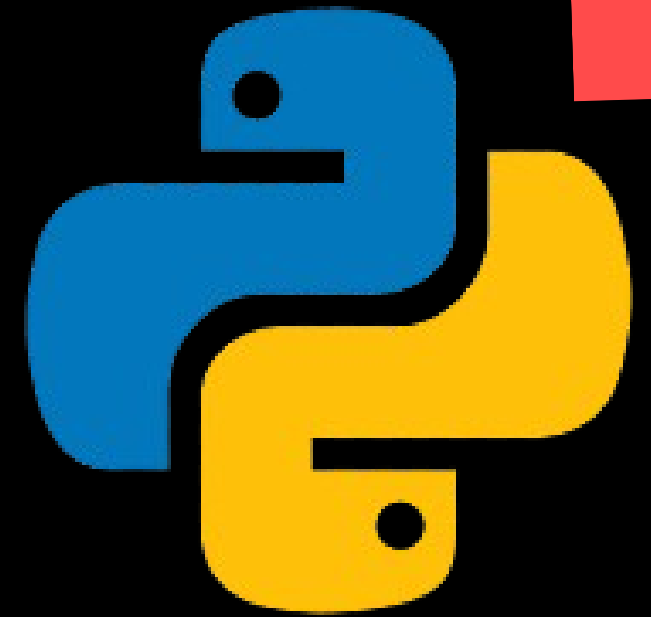
Talk with your expense bills!



#4 Data extraction from extracted text

> Extract column information like invoice_no, amount, description, etc.

> LLM does not understand the python dictionary

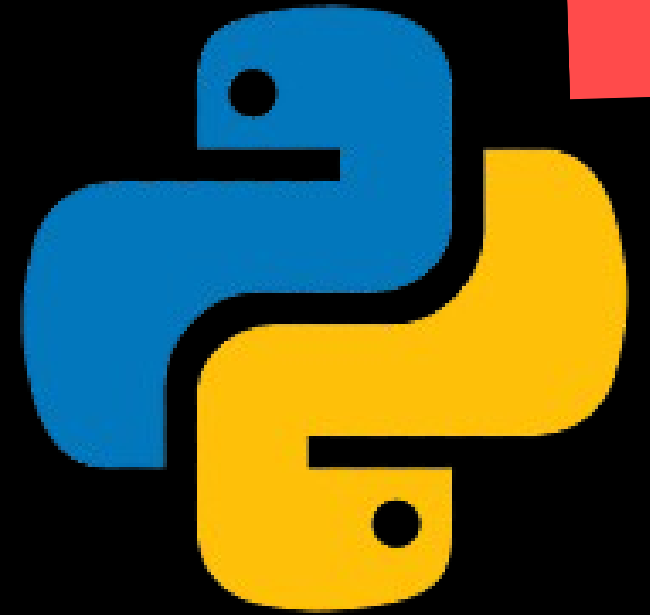


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#5 Categorization of expense

> Based on broad categories such as food, utilities, grocery expenses, travel, etc :
categorizing the expense into different category

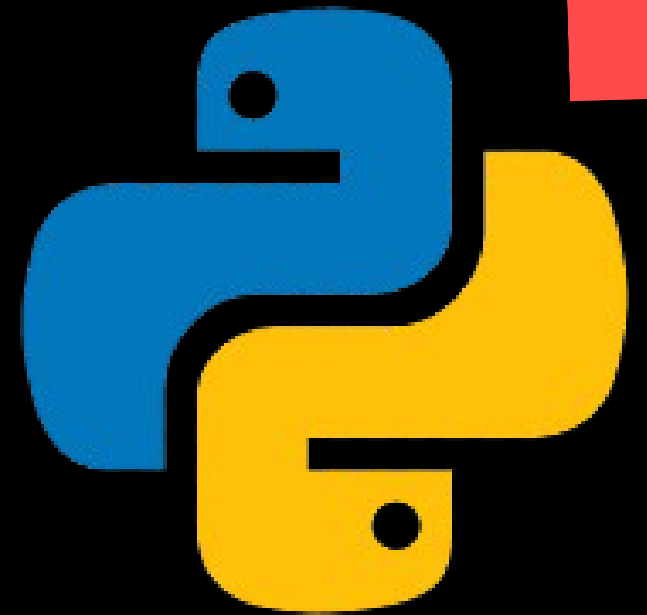


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#6 Creating table in sqlite3

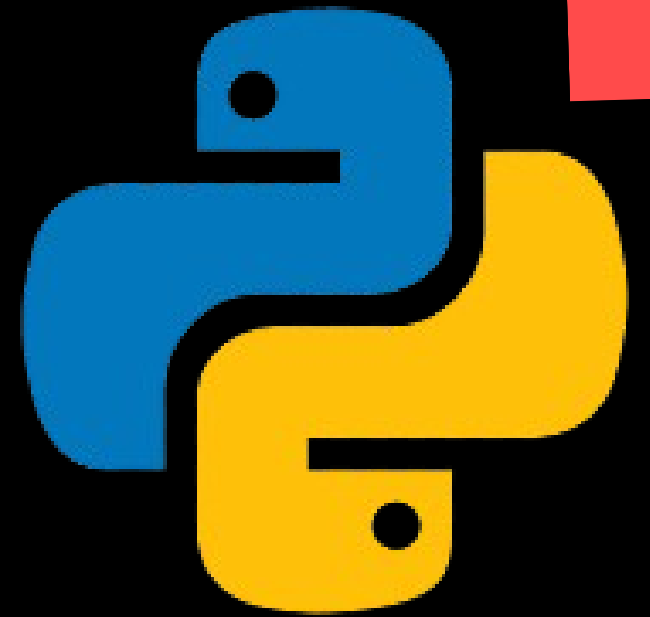
> Creating table with column headers for storing information from bills



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#7 parser - extracting information into dictionary format



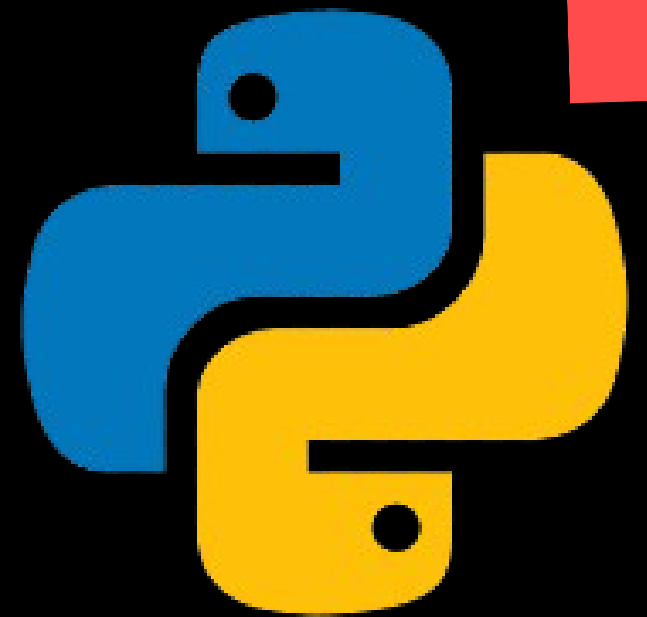
The Python logo, consisting of two interlocking snakes, one blue and one yellow, set against a black background.

- i. Connecting with table**
- ii. breaking description into line items**
- iii. uploading data into table**

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> frontend and Creating chat box



Talk with your expense bills!

